DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Computer Science and Engineering) SEMESTER - 5 Winter 2025 (Regular)

Course :Bachelor of Technology (Computer Science and Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTCOE504B - NUMERICAL METHODS

Time : 3 Hours] **[Total Marks : 60**

Instructions to the Students:

1.Each question carries 12 marks.
2.Question No. 1 will be compulsory and include objective-type questions.
3.Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4.Use of non-programmable scientific calculators is allowed.
5.Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question) 12

1 Interpolation is the process of estimating values
a) Outside the given data range b) At random points
c) Using derivatives only d) Within the given data range

2 Picard's method is based on
a) Differentiation b) Integration c) Finite differences d) Matrix inversion

3 The Relaxation method improves convergence by introducing
a) Pivot element b) Determinant c) Step size d) Relaxation factor

4 Compared to the trapezoidal rule, Simpson's rules generally give
a) Same accuracy b) Lower accuracy
c) Zero error d) Higher accuracy

5 Newton's forward interpolation formula is suitable when the value lies
a) Near the end of the table b) Near the beginning of the table
c) Outside the table d) At the midpoint

6 In the bisection method, the root always lies in an interval where
a) $f(a)f(b)<0$ b) $f(a)=f(b)$ c) $f'(x)=0$ d) $f(x)$ is linear

7 Gauss–Jordan method directly transforms the matrix into
a) Upper triangular form b) Diagonal form
c) Sparse form d) Lower triangular form

8 The Runge–Kutta fourth–order method evaluates the slope
a) Once b) Twice c) Three times d) Four times

9 The backward difference of a function is defined as
a) $f(x)-f(x-h)$ b) $f(x+h)-f(x)$ c) $f(x+h)-f(x-h)$ d) $hf'(x)$

10 Newton–Cotes formulas are obtained using
a) Polynomial interpolation b) Finite differences
c) Differential equations d) Iterative methods

11 Euler's method is a ______ numerical method.
a) Implicit b) Explicit c) Multi-step d) Predictor–corrector

12 Simpson's one-third rule requires the number of subintervals to be

a) Odd b) Prime c) Even d) Multiple of three

Q2. Solve the following.

A) Find the root of the equation x3 – 4x – 9 = 0, using the bisection method, correct to three decimal places. 6

B) Find by Newton's method, the root of the $xe^x - 2 = 0$ equations correct to three decimal places. 6

Q3. Solve the following.

A) Solve by the Gauss-elimination method. 6

$10x - 7y + 3z + 5u = 6$; $-6x + 8y - z - 4u = 5$;

$3x + y + 4z + 11u = 2$; $5x - 9y - 2z + 4u = 7$.

B) Solve the following equations by the Gauss-Jordan method: 6

$2x - 3y + z = -1$; $x + 4y + 5z = 25$; $3x - 4y + z = 2$

Q4. Solve Any Two of the following.

A) In the following table, the values of y are consecutive terms of a series, of which 12.5 is the 5th term. Find the first and tenth terms of the series. 6

x:	3	4	5	6	7	8	9
y:	2.7	6.4	12.5	21.6	34.3	51.2	72.9

B) Evaluate: 6

a) $\Delta\,(x + \cos x)$ b) $\Delta^2 \sin(px + q)$ c) $\Delta(e^{3x} \log 2x)$

C) Estimate from the following table f(3.8) to three significant figures using Newton backward interpolation formula: 6

x:	0	1	2	3	4
y:	1	1.5	2.2	3.1	4.6

Q5. Solve Any Two of the following.

A) Derive the formula for Simpson's 1/3rd rule from Newton Cotes Quadrature formula and evaluate $\int_0^1 \frac{dx}{x^3+x+1}$ using Simpson's 1/3rd rule, choosing a step length of 0.25. 6

B) Estimate from the following table f(3.8) to three significant figures using Newton backward interpolation formula: 6

x:	0	1	2	3	4
y:	1	1.5	2.2	3.1	4.6

C) Derive the formula for Simpson's Three-Eighth Rule from Newton – Cote's quadrature formula. 6

Q6. Solve Any Two of the following.

A) Using the Runge-Kutta method of fourth order, solve $\frac{dy}{dx} = \frac{y^2 - x^2}{y^2 + x^2}$ with $y(0) = 1$ at x= 0.2 and x = 0.4. 6

B) Using Euler's method, solve for y at $x = 0.1$ from $dy/dx = x + y + xy$, $y(0) = 1$, taking step size $h = 0.025$. 6

C) Solve the following by Euler's modified method at x = 1.2 and 1.4 with h = 0.2. 6

$$\frac{dy}{dx} = \log(x+y),\ y(0) = 2$$

*** * * End * * ***